

Tutorial 3: Watermarking and Mechanistic Interpretability

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Instructions

- Attempt all exercises before the tutorial session.
- Discuss your reasoning, not just the final answer.
- Collaboration is encouraged, but think of your own solutions first.

Exercise 1. Laura was asked to construct an algorithm that would “watermark” large language model text, so text generated by an LLM can later be automatically identified by another verification algorithm. Laura thought about the following algorithm. Given a language model, the algorithm works by apriori dividing its vocabulary set into two subsets of *roughly equal* probability (0.5) by estimating a word probability based on counts from a large corpus. The two sets are: one set of “green” tokens and one set of “red” tokens. Then, given a prompt, the LLM is constrained to only generate text using the green tokens. Following that modification to the LLM, Laura’s idea was to test, for any text, whether it uses green tokens only. If it does, then we can classify it as generated by the LLM.

1. Explain in two sentences one major disadvantage of this algorithm. Will it work in practice?
2. Laura further decided to improve her algorithm, and rather than dividing the vocabulary set apriori, her algorithm now divides the vocabulary set into a set of green and red tokens per *prompt*, randomly as in her original algorithm, independently of the prompt. What is the major issue with this new modification?
3. Suggest an alternative that you expect would work well, and overcome both the first disadvantage and the second disadvantage in the above subquestions (a) and (b). You can base your answer on an existing algorithm described in class.
4. Note briefly one disadvantage of the algorithm you proposed.

Solution.

1. A major disadvantage is that certain words will be blocked and this will remove the possibility of possibly using important tokens completely. (To tutors: reinforce the algorithm and explain it again in rough sketch.)
2. The algorithm is problematic because the randomisation into green/red tokens is not dependent on the prompt or anything fixed, so there is no way to recover the buckets. (To tutors: keep an open mind about solutions to this question, as there might be other explanations. Create a discussion.)
3. Set a hash code per token generated, initialise the pseudo random number generator, and split based on that. This means we can regenerate the green/red tokens based on the text. Alternatively, but not so great (only half of the marks): set the pseudo random number generator based on the prompt (prompt is not necessarily included in the text). (To tutors: same here, create a discussion.)

4. Low-entropy states are still problematic. (To tutors: Same here, there are other issues that might be mentioned, not necessarily related to the specific setup. For example, using third-party software or LLM to paraphrase the outputs.)

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Exercise 2. This question gives a treatment to sparse autoencoders (SAEs). Nick wanted to test his LLM, which was pretrained previously on a large corpus of text, with SAEs. He took a corpus of multilingual Wikipedia text, and ran it through the LLM. He then recorded the activations from the residual stream, and then eventually trained an SAE on these activations. The dimension of his representation was 768 while the SAE had a layer of dimension 20,000.

1. What does an SAE architecturally look like? What does it take as input, what is the middle layer, and what is the output?
2. Nick noted that when he tested which features the SAE has, he could find a much larger number of features firing in the SAE sparse layer (overall) than the dimension of his representations. How could that be?
3. Nick noted three interesting sparse features:
 - Feature 4,678 - which fires whenever male pronouns and stereotypically-male adjectives appeared in the original input of the activation;
 - Feature 5,021 - which fires whenever female pronouns and stereotypically-female adjectives appeared in the original input of the activation;
 - Feature 9,312 - which fires whenever any pronoun appeared in the original input of the activation.

Nick wants to build a classifier, which based on a one-sentence summary of a person's biography, decides on the potential profession of the person to whom the biography belongs. How would you construct such a classifier using the LLM? What would you suggest Nick to do with his classifier to be less based on stereotypical associations of professions with genders? Naturally, you should consider using an SAE in the process.

4. How could Nick quantitatively measure how much his classifier relies on stereotypical professional features before and after his intervention?

Solution.

1. An SAE has the input layer, where the activations come in, a middle sparse very wide layer which fires the features, and finally a reconstruction layer that attempts, based on the feature, to reconstruct the activation. (To tutors: You can give more intuition based on the equations provided in Lecture 16, slide 22; go over these equations; create a discussion around SAEs and explain them).
2. This is the result of superposition. The 768 dimensional space contains complex features which are not necessarily orthogonal. The SAE is able to separate this superposition ("invert the superposition") so that these features are explicit.
3. This is meant to be an open-ended question about a simple version of debiasing. To build the classifier using the LLM, Nick can run a one-sentence summary of the biography through the LLM to get an embedding, and then use a classifier layer to classify into a profession based on existing training data (through finetuning). What Nick can do is

replace the activations with the SAE reconstructed activations. However, the SAE would be changed (in the middle layer) such that the two features that are associated with specific genders are further deactivated, and whenever one of them is activated, feature 9,312 is activated instead. This is also referred to as “steering”, and would could partially neutralise the reliance on gender to predict the profession. Create a discussion here: note that this will not *necessarily* improve predictive power of gender; however, it will make prediction more fair in the sense of relying less on potentially stereotypical gender cues rather than actual profession information.

More technical discussion can include how to train the classifier. Where do we get the data from? We mentioned that the SAE was trained on Wikipedia text. We can take, for example, parts of Wikipedia that provide biographies rather than arbitrary entries in Wikipedia to train both the classifier and the SAE. (To tutors: you can mention that indeed a well-known dataset in NLP was developed that way, called BiasBios, where there was an attempt to predict *professions* based on biographies without taking into account gender information.)

4. Basically, compare predictions with / without feature suppression and see how big the difference is.

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